

High Performance Computing with GPUs (CS4110)

Course Instructor(s):

Dr. Imran Ashraf

Section(s): A, B

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 43

Total Questions: 6

Date: Feb 25, 2025

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

1. Solve all questions and all parts of each question in order.
2. The source code should be properly indented.
3. Simply providing the answer is not enough to get the marks. You need to justify your answers.
4. Write the question number as well as the part number of each question.

[CLO 4: Analyze the performance of an application running on an HPC system to improve compute and memory performance.]

Q1:

[4+4+4 marks]

A scientific simulation runs on a **CPU-only system** in **100 seconds**. A portion **P** of the program can be offloaded to a **GPU**, while the remaining **(1 - P)** must still be executed on the **CPU**. The GPU can execute the parallel portion **50× faster** than the CPU. In the light of these details, answer the following questions.

- A) If **60% of the program can be offloaded to the GPU**, calculate the **overall speedup**.
- B) What percentage of the program **must be offloaded to the GPU** to achieve at least a **20× speedup**?
- C) Suppose **kernel launch overhead and memory transfers** introduce an **additional 5% execution time overhead** in the parallel portion. How does this impact the expected speedup?

[CLO 3: Develop data-parallel solutions using appropriate programming models.]

Q2:

[4+4 marks]

You are developing a CUDA program which squares each element of the input array. For an **array of size N**, where $N = 1000000$ and the block size is 256,

- A) Provide CUDA code for kernel launch configuration (ONLY) that ensures all elements are processed.
- B) Provide the complete kernel. Inside the kernel, prevent **out-of-bounds** access.

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[CLO 3: Develop data-parallel solutions using appropriate programming models.]

Q3:

[1+2 marks]

Given the kernel launch:

```
myKernel<<<64, 512>>>(d_data);
```

- A) How many total threads are launched?
- B) Modify the launch configuration to maintain the same total threads but with 256 threads per block instead of 512.

[CLO 3: Develop data-parallel solutions using appropriate programming models.]

Q4:

[4 marks]

A CUDA kernel is launched with:

```
dim3 blockSize(16, 16);
```

```
dim3 gridSize(32, 32);
```

```
myKernel<<<gridSize, blockSize>>>(d_data);
```

How would you compute a **unique 2D global index** (global_x, global_y) for each thread inside the kernel?

[CLO 3: Develop data-parallel solutions using appropriate programming models.]

Q5:

[10 marks]

Given the following host side of CUDA program. Point out the error(s) and fix them.

```
int main() {  
    int N = 1000;  
    int * h_array, * d_array;  
  
    // Allocate memory on device  
    cudaMalloc(d_array, N * sizeof(int));  
  
    // Allocate memory on host  
    h_array = (int * ) malloc(N * sizeof(int));  
  
    // Copy data from host to device  
    cudaMemcpy(h_array, d_array, N * sizeof(int), cudaMemcpyDeviceToHost);  
  
    // Initialize host array
```

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```
for (int i = 0; i < N; i++) {  
    h_array[i] = i;  
}  
  
// Launch kernel  
kernel << < (N / 256), 256 >>> (d_array, N);  
  
// Copy results back to host  
cudaMemcpy(d_array, h_array, N * sizeof(int), cudaMemcpyHostToDevice);  
  
// Free memory  
cudaFree(d_array);  
free(h_array);  
  
return 0;  
}
```

[CLO 1: Describe the terms related to HPC systems, GPU architecture and GPU programming models.]

Q6:

[3+3 marks]

Compare and contrast the following by providing at least 1 example for each.

- A)** Homogeneous vs Heterogeneous computing
- B)** Grid dimensions vs Block dimensions